



**TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC 2/ IP YEAR 6 2017**

CANDIDATE
NAME

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H2 BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

**Thursday 14 September 2017
2 hours**

Additional materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Section A

Write your name, Centre Number, index number and class in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graph.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the separate Writing Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Q1	/11
Q2	/11
Q3	/28
Essay	/25
Total	/75

This document consists of **17** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows an example of how PCR and gel electrophoresis can be used to identify the allele that codes for the blood type antigen that is present in an individual.

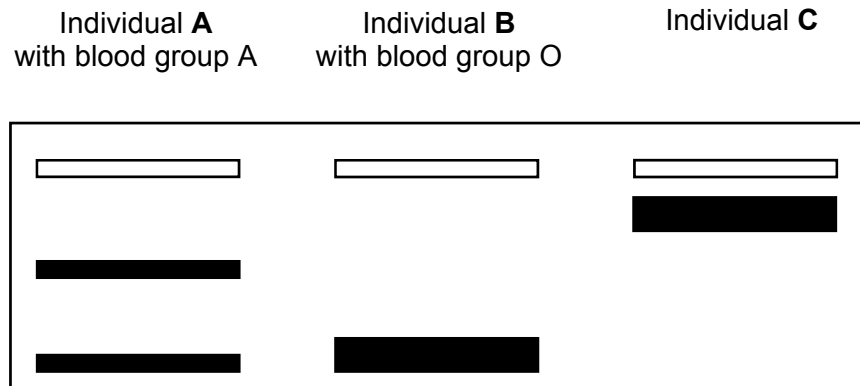


Fig. 1.1

- (a) State if individual **C** can donate blood to individual **A**. Explain your answer. [2]
1. **Individual C cannot donate blood to individual A.**
 2. **Individual A has anti-B antibodies that will bind to the antigen B on the surface of the RBC of Individual C while Individual C has anti-A antibodies that will bind to the antigen A on the surface of the red blood cell of Individual**
 3. **and cause agglutination/ haemolysis.**
- (b) State **one** limitation of using PCR in determining the genotype of the individual. [1]
- ANY ONE:**
1. **The DNA sequence must be known in order to design primers for successful amplification.**
 2. **If the sample is contaminated, amplification of unwanted DNA may also take place.**
- (c) Explain why Southern blotting can be used to confirm the presence of the allele coding for blood type A antigen. [2]
1. **The probes are radioactively labelled**
 2. **and only hybridize via complementary base pairing with**
 3. **the the allele that codes for blood type A antigen**
 4. **The probe can then be detected as a darkened band using autoradiography.**

Malaria is a vector-borne disease caused by *Plasmodium* spp.

Plasmodium gametocytes can differentiate into female and male gametes. To correctly identify male and female *Plasmodium* gametes, real-time polymerase chain reaction (PCR) was used. 60ng of DNA samples from different types of *Plasmodium* gametes were first prepared in separate tubes. Primers for a specific gene sequence were added to each tube of DNA sample. Free nucleotides tagged with fluorescent dye were also added.

Fig. 1.2 shows the level of fluorescence measured as PCR progressed.

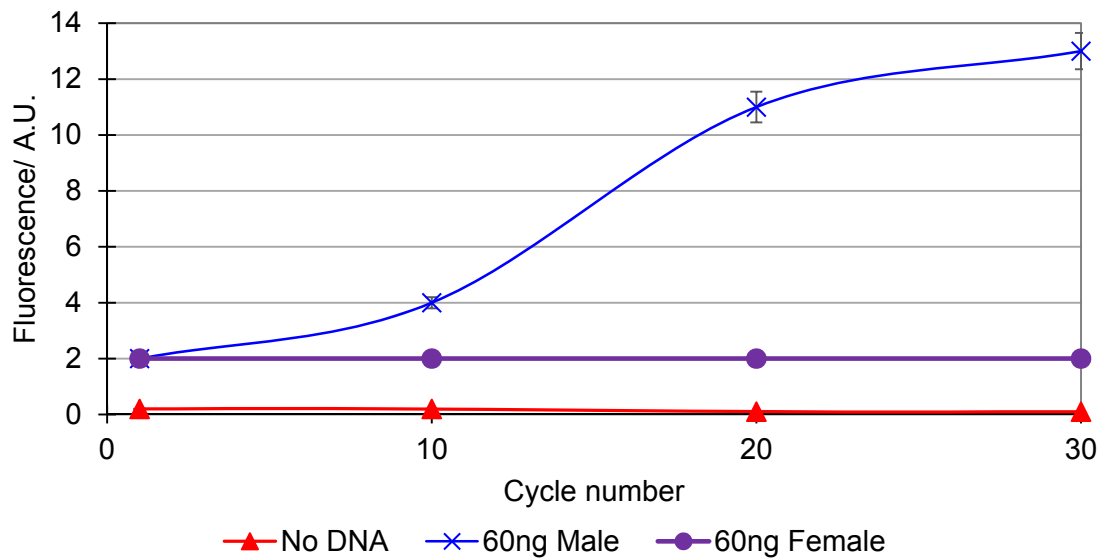


Fig. 1.2

- (d) (i) State the type of chromosome on which the gene is found on. Explain your answer. [2]

1. Y chromosome

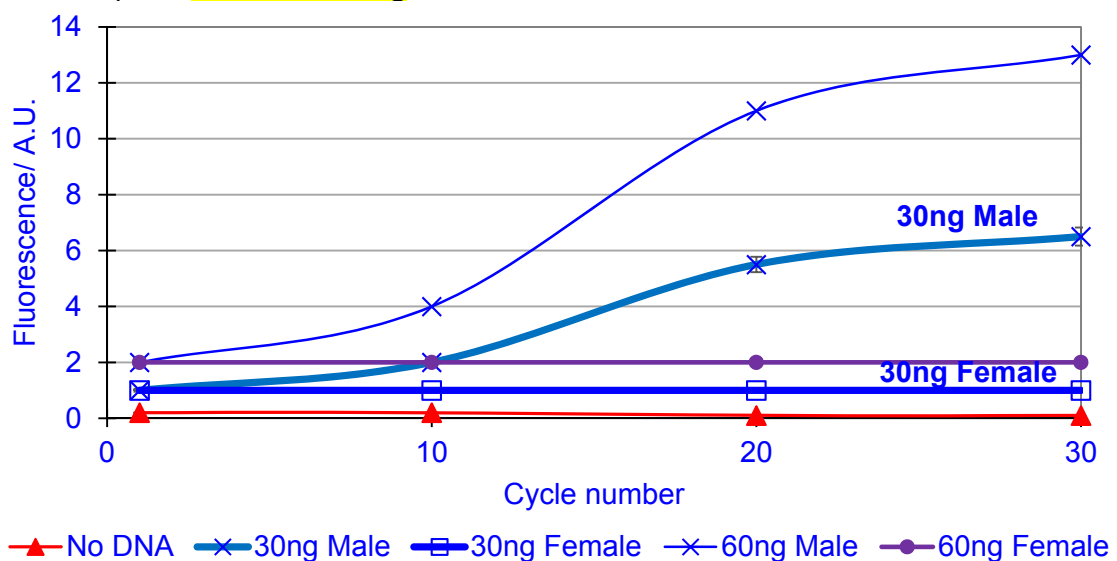
2. The primers only bind and amplifies the gene in the Y chromosome, resulting in the increase of fluorescence level 2A.U. at cycle 1 to 13A.U. at cycle 30

- (ii) Predict and explain what will happen to the fluorescence level after 30 cycles for the DNA sample from the **male** gametes. [1]

1. It will remain constant as the primers/ nucleotides have been used up.

- (iii) Real-time PCR can also be used to determine the initial quantity of the sample.

On Fig. 1.2, sketch how the graph will look like when the mass of starting DNA sample is **reduced to 30ng**. [1]



1. Half of the values throughout, starting from 1 A.U.

- (e) Distinguish the products formed from real-time PCR with that of DNA replication. [2]

ANY ONE:

PCR	DNA replication
Only the <u>region flanked by the primers</u>	<u>Entire genome</u>
<u>High</u> likelihood of having mutation in the sequences	<u>Low</u> likelihood of having mutation in the sequences
<u>DNA primers become part of the newly synthesized strand</u>	<u>RNA primers are removed and replaced by DNA nucleotides</u>

[Total: 11]

2 Hepatitis C virus (HCV) is an enveloped, positive single-stranded RNA virus that infects hepatocytes (liver cells). The virus is attached to a low-density lipoprotein (LDL) upon its release from hepatocytes.

Fig. 2.1 shows how HCV enters and leaves a hepatocyte.

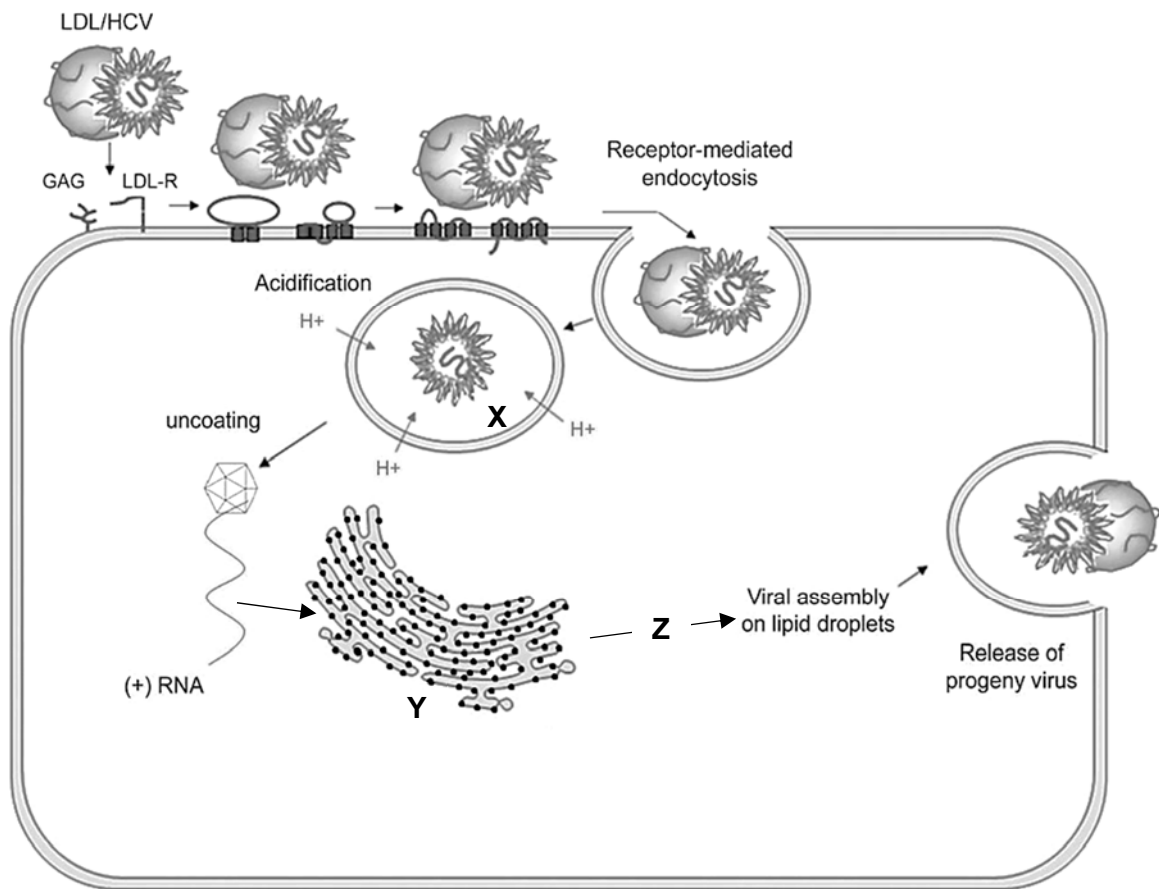


Fig. 2.1

(a) With reference to Fig. 2.1, state the role of LDL with respect to the viral replication cycle. [1]

- As a ligand, to bind to the receptors on surface of hepatocytes to trigger receptor mediated endocytosis.

(b) Describe the functions of organelle X with respect to the replication cycle of HCV. [2]

1. Once HCV-LDL complex is endocytosed within organelle X, the pH decreases and triggers the removal of the envelope
2. Thus releasing the nucleocapsid into the cytoplasm of the hepatocyte.

- (c) One of the organelles involved in HCV replication cycle is **Z**.
Suggest the identity of **Z** and its role in the virus replication pathway. [2]

Z: Golgi apparatus.

- **To chemically modify, sort and package the viral proteins**

- (d) Explain the role of the positive single-stranded RNA as a genetic material in HCV. [2]

1. **to synthesize (-)RNA strand → synthesis of more (+)RNA strands.**
2. **to synthesize viral proteins.**

The HCV core protein has been found to be oncogenic. An increase in the viral core proteins within hepatocytes can lead to the activation of cell proliferation signaling pathway.

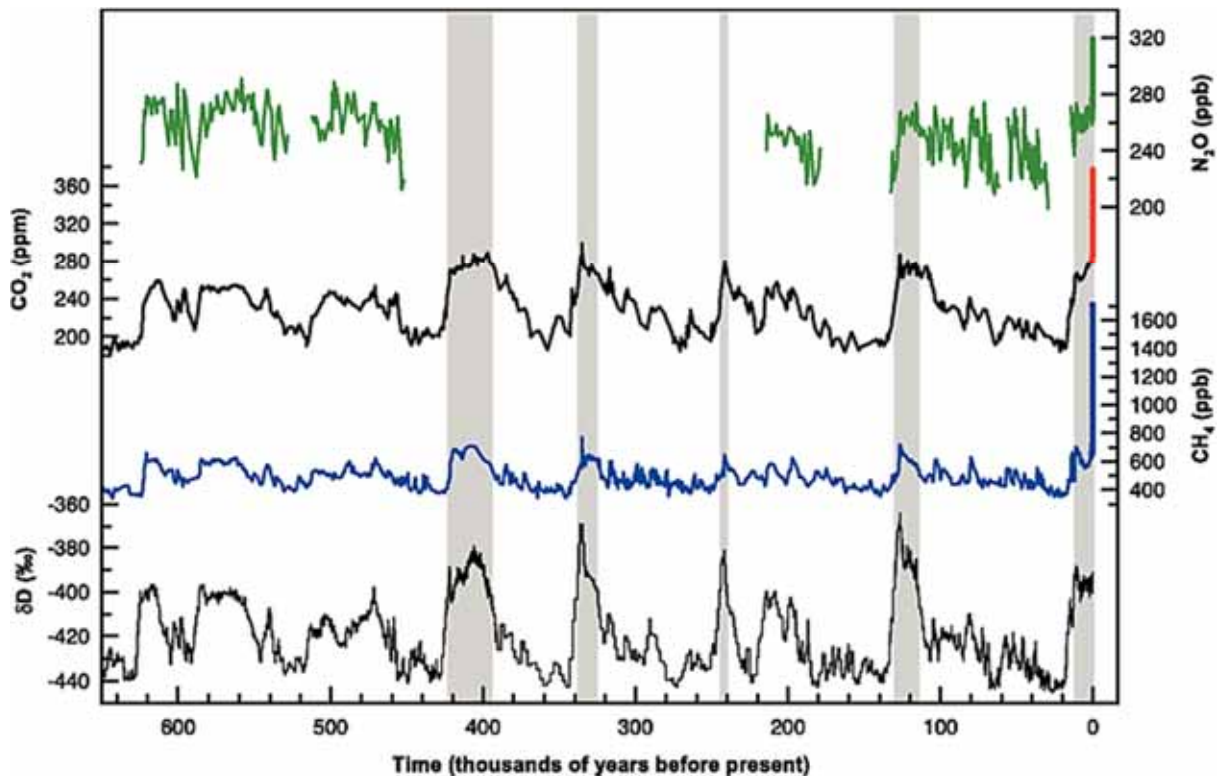


Fig. 2.2

- (e) Define the term *oncogenic* in this context. [2]
- **It leads to constitutive activation of cell proliferation signalling pathway, thus leading to uncontrolled cell division.**
- (f) Out of 3 million individuals infected with HCV, less than 5% developed liver cancer. Account for the low percentage of HCV infected individuals developing liver cancer. [2]
1. **Half a dozen independent mutations that must be present in a single cell**
- Any one:**
2. **The mutation or loss of several tumour suppressor genes. And mutations need to knock out both alleles in a cell's genome to block tumour suppression.**
- OR**
3. **gene for telomerase is activated.**
- OR**
4. **Angiogenesis, metastasis**

[Total: 11]

- 3 Fig. 3.1 shows the atmospheric concentrations of carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O) in the air that was trapped within ice cores. The changes in temperature is reflected as variation of deuterium (δD). The shading indicates the last interglacial warm periods.



Source: IPCC Fourth Assessment Report: Climate Change 2007

Fig. 3.1

- (a) (i) Without quoting any numerical data, state the general relationship between the greenhouse gases and temperature in Antarctica. [1]
1. The concentration of greenhouse gases is proportional to the temperature in Antarctica,
 2. and they have similar peaks that coincide at the same time
- (ii) Suggest the possible consequences of increased temperature on the physical environment in Antarctica. [2]
- ANY TWO:**
1. Arctic sea ice cover will continue to shrink and thin
 2. Global glacier volume will decrease and
 3. Permafrost will continue to thaw
 4. Increase in sea level
 5. Decrease in land mass

Climate change critics argue that the climatic variation as seen in Fig. 3.1 is natural.

Fig. 3.2 shows how carbon dioxide concentration and temperature has changed since year 1000, which may be used to convince critics that the extent of climate change over the last millennium is beyond what was previously seen.

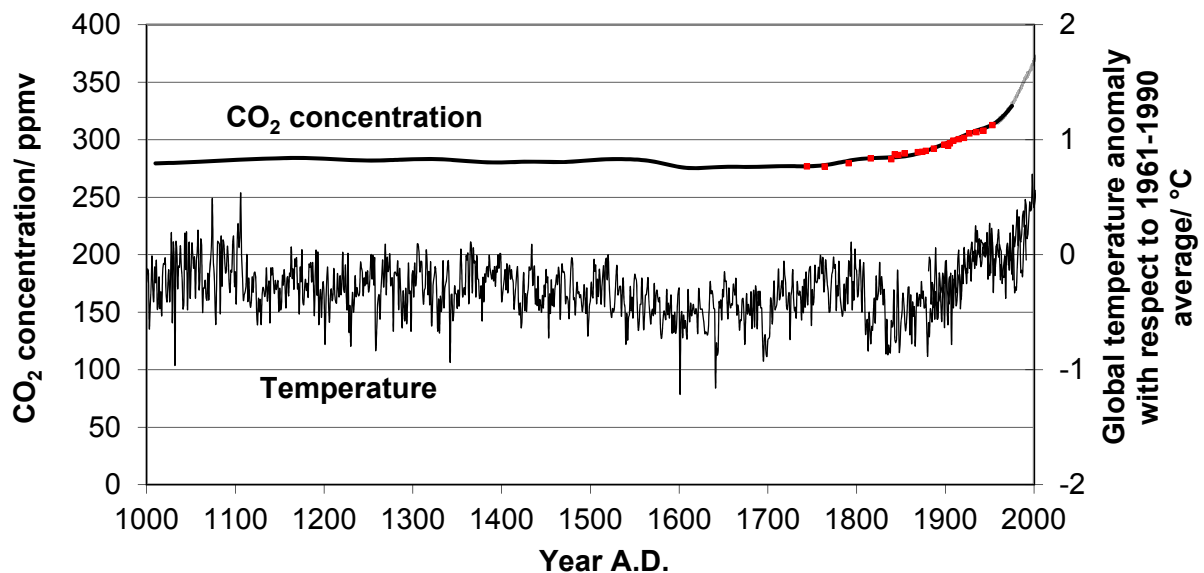


Fig. 3.2

(iii) Account for the trend of carbon dioxide concentration. [3]

1. relatively constant at 280ppmv from 1015 to 1800A.D..
2. increases steeply from 280ppmv in 1800A.D. to 375ppmv in 2000A.D..
3. more carbon dioxide is released than it is being removed
due to:
4. Industrial Revolution/ Industrialization; Fossil fuel combustion to produce electricity/ transport ; Deforestation/ Land use change for urban development or agriculture; Agriculture and food production/ Rearing husbandry/ Consumption of meat

Climate change affects plants. Plants can be categorized based on the way they photosynthesize. Most plants are C3 plants because their first photosynthetic product is a three carbon compound. Examples of C3 plants include barley, oats and wheat commonly grown in temperate regions. On the other hand, common grass crops of tropical regions, such as maize, sorghum and sugarcane, are C4 plants.

The rate of photosynthesis for both temperate and tropical plants were measured and shown in Fig. 3.3.

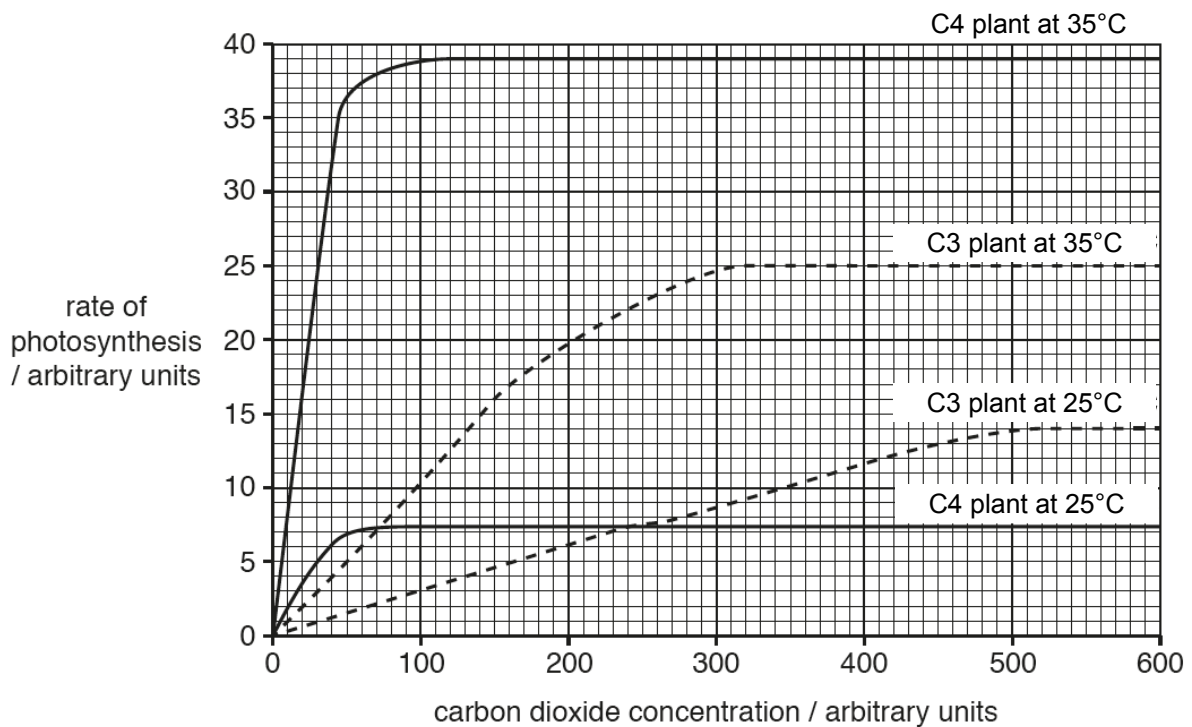


Fig. 3.3

(b) Using evidence from Fig. 3.3, suggest which type of plant is better adapted to the impacts of climate change. [2]

1. C4 plants
2. At high carbon dioxide concentration and high temperature, the rate of photosynthesis for C4 plants at 39A.U. is HIGHER than that of C3 plants at 25°C.

Fig. 3.4 shows the data collected on the distribution of C4 plants over time, whereby the local temperature was shown to have increased since 1940s. Three populations of the same species of C4 plants were tracked over time.

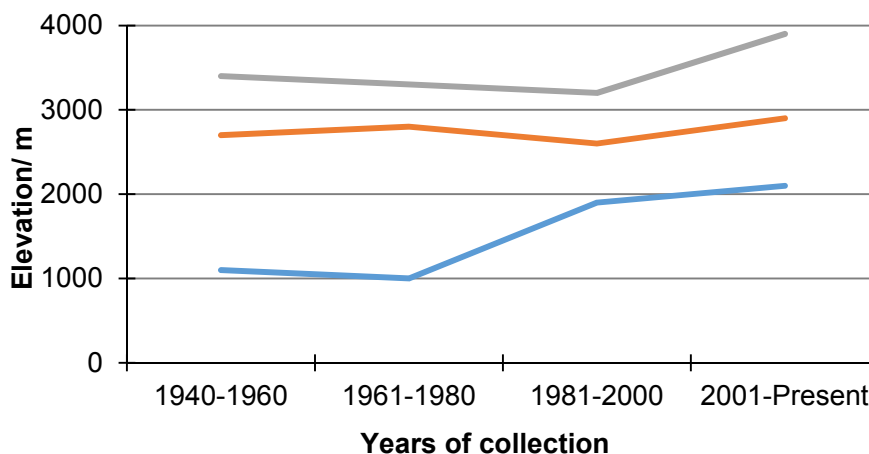


Fig. 3.4

(c) Account for the effect of increased temperature on the distribution (altitude) of C4 plants. [2]

1. increases from the range of 1100 to 3400m in 1940s to 2100 to 3900m presently
2. as temperature at higher altitude is cooler thus favorable
3. for the C4 plants' growth

Fig. 3.5 shows the effects of climate change on the potential change in yield and geographical distribution of wheat and maize plants at sub-continental level from 2005 to 2050.

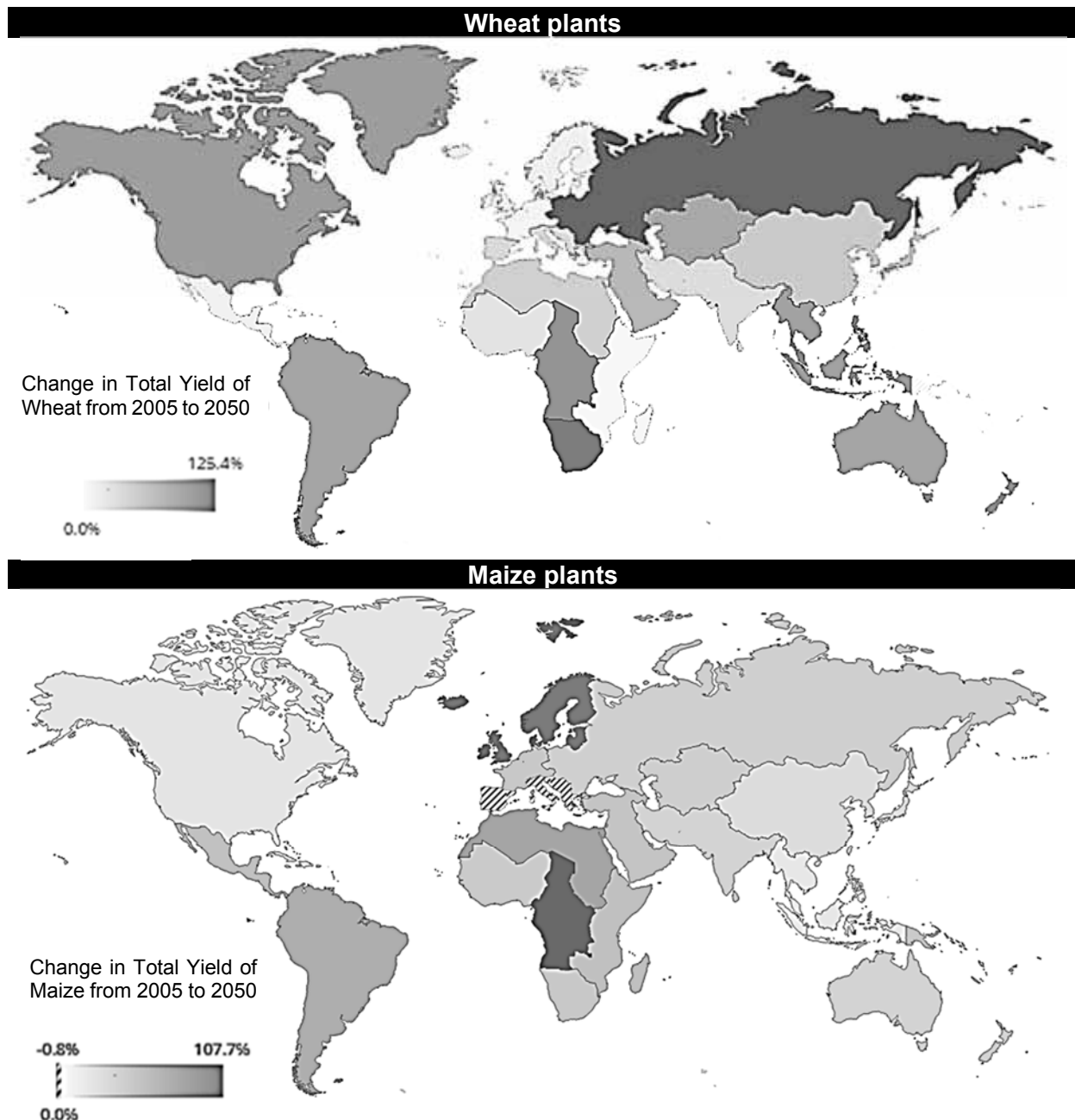


Fig. 3.5

- (d) State the possible consequence of increased temperature on global wheat and maize supply in 2050. [1]

1. Total yield of wheat and maize increase
2. total yield of wheat is greater than that of maize.

Climate change also has impact on the physiology of insects, thus affecting the transmission of vector-borne diseases. One such vector is the mosquito.

- (e) (i) Outline the general life cycle of a mosquito. [2]

1. The female mosquitoes lay eggs in stagnant water
2. develop as larvae in water,
3. develop into PUPAE in water,
4. mature/ develop into an adult mosquito.

- (ii) Blood meals are a good source of protein for mosquitoes for the production of eggs.

Explain why blood is a good source of protein.

[1]

1. Blood plasma contains many proteins.
2. Red blood cells contain haemoglobin/ plasma proteins which is
3. required for the growth of the mosquito.

Fig. 3.6 shows the effect of temperature on the egg-to-adult survival rates and the behaviour of adult mosquitoes.

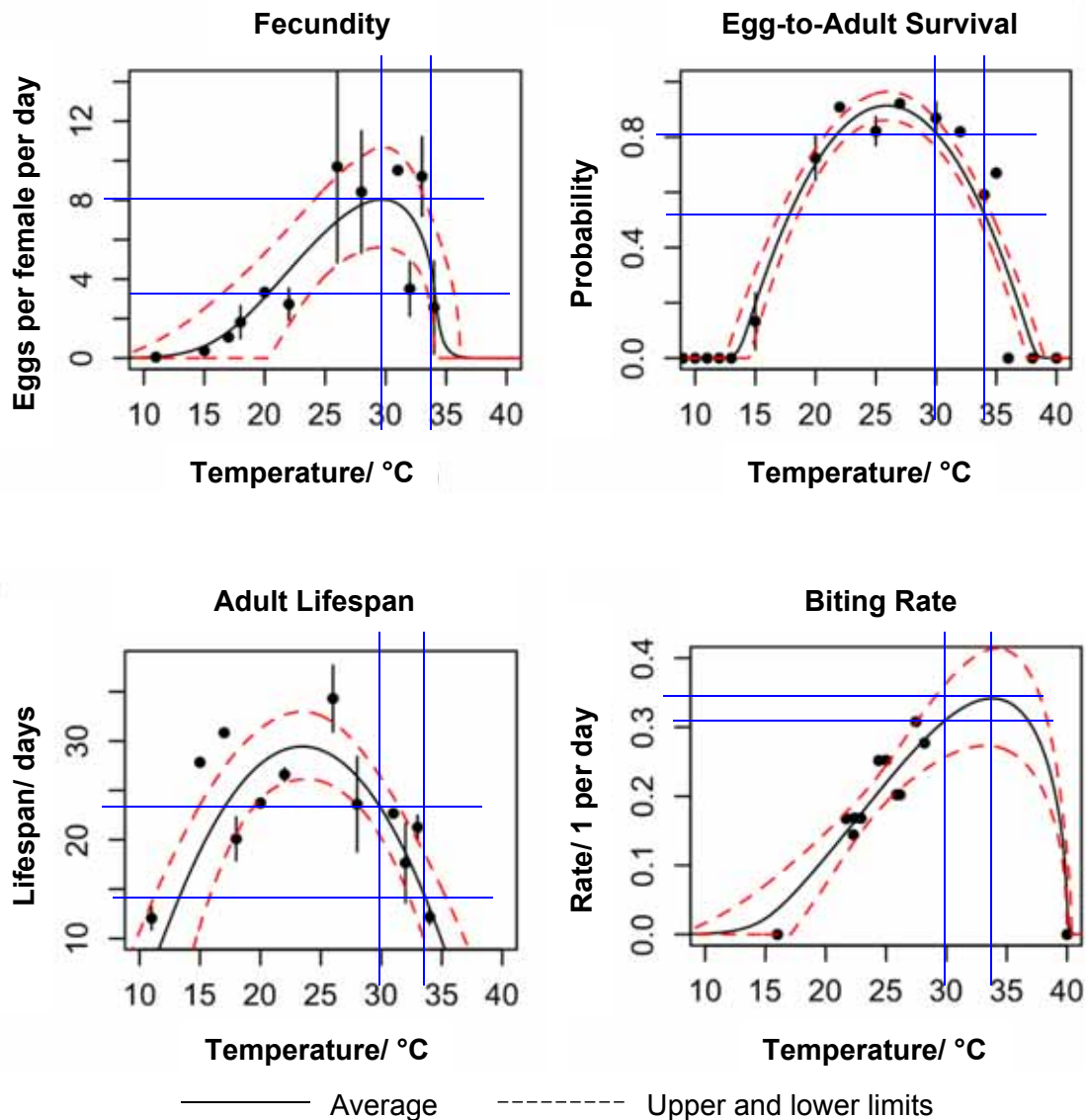


Fig. 3.6

- (f) Using evidence from Fig. 3.6, discuss the rate of transmission of diseases caused by mosquitoes when temperature increases from 30 to 34°C.

[3]

1. As temperature increases from 30°C to 34°C, the AVERAGE fecundity decreases from 8 to 3 eggs laid per female per day,
2. the AVERAGE egg-to-adult survival rate decreases from probability of 0.8 to 0.5,
3. the AVERAGE lifespan of adult mosquitoes decreases from 23 to 14 days.

4. Thus, there will be fewer mosquitoes in the environment,
5. thus the OVERALL possibility of transmission of mosquito diseases is likely to decrease,
6. even though the AVERAGE biting rate increases from 0.31 to 0.34 per 1 (mosquito) per day.

Table 3.1 describes the two types of mosquitoes that can transmit dengue.

Table 3.1

<i>Aedes aegypti</i> (Yellow fever mosquito)	<i>Aedes albopictus</i> (Asian tiger mosquito)
Found both indoors and outdoors	Found mostly outdoors (e.g. gardens)
High preference for taking blood meals from humans and to lesser extent from domestic mammals, which makes it a very capable vector of dengue viruses	Bites humans but also a variety of available domestic and wild vertebrates that do not carry the dengue viruses, which lowers its capacity to transmit them

- (g) Explain why these *Aedes* mosquitoes are considered two species. [3]
1. They are classified based on ecological species concept as the *Aedes* mosquitoes live in different niches
 2. *Aedes aegypti* : both indoors and outdoors,
 3. *Aedes albopictus* : outdoors
 4. Source of food: *Aedes aegypti* preferentially take blood meals from humans whereas *Aedes albopictus* takes blood meals from humans and vertebrates.
 5. They do NOT interbreed/ mate to produce fertile, viable offspring.
- (h) Outline how the dengue virus infects immune cells in humans and spreads in the body. [2]
1. The virus infects immature Langerhans cells and keratinocytes and enters the cell via receptor-mediated endocytosis.
 2. The new dengue virus exits via exocytosis with the help of host protease furin.

In October 2016, the Health Sciences Authority (HSA) approved the use of Dengvaxia, the world's first dengue vaccine, in Singapore. The vaccine was recommended for people aged 12 to 45.

- (i) Explain why the vaccine was not effective for people aged 46 and above. [2]
1. As people age, the thymus shrinks,
 2. repertoire of naïve T cells will be lower.
 3. Activation of T cells and B cells by the vaccine will be lower,
 4. Unable to form memory T and B cells

Wolbachia is a diverse group of naturally occurring intracellular bacteria that infect and live in invertebrate cells and manipulate the biology of their hosts. Although they are found in more than 60 per cent of insect species, they are not found in the main vector of dengue *Aedes aegypti* in Singapore.

In order to suppress the mosquito population, researchers in Singapore conducted a field study that released fertile *Wolbachia*-carrying male mosquitoes into the urban built environment. This could greatly reduce the numbers of mosquitoes as the resulting zygote do not hatch.

Fig. 3.7 shows one possible effect of *Wolbachia* infection on the first zygotic mitosis.

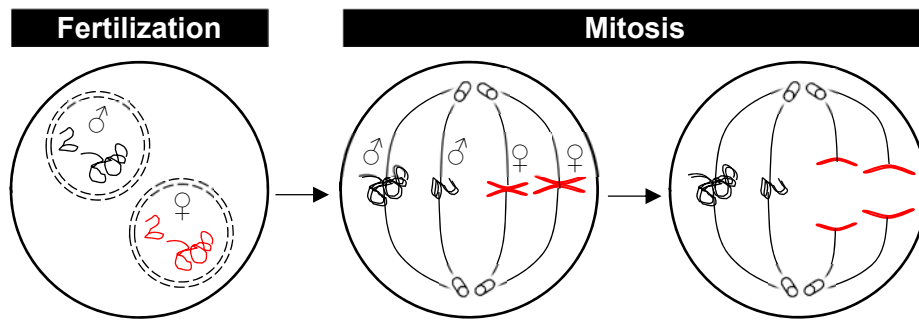


Fig. 3.7

- (j) (i) Explain why *Wolbachia* infection will be successful in reducing the number of mosquitoes over time. [2]
1. The paternal chromosomes of the zygote
 2. do not condense during prophase,
 3. and thus they cannot separate during anaphase.
 4. results in non-disjunction, mitosis is arrested / cannot form daughter cells.

Fig. 3.9 shows the food web in which mosquitoes are found in.

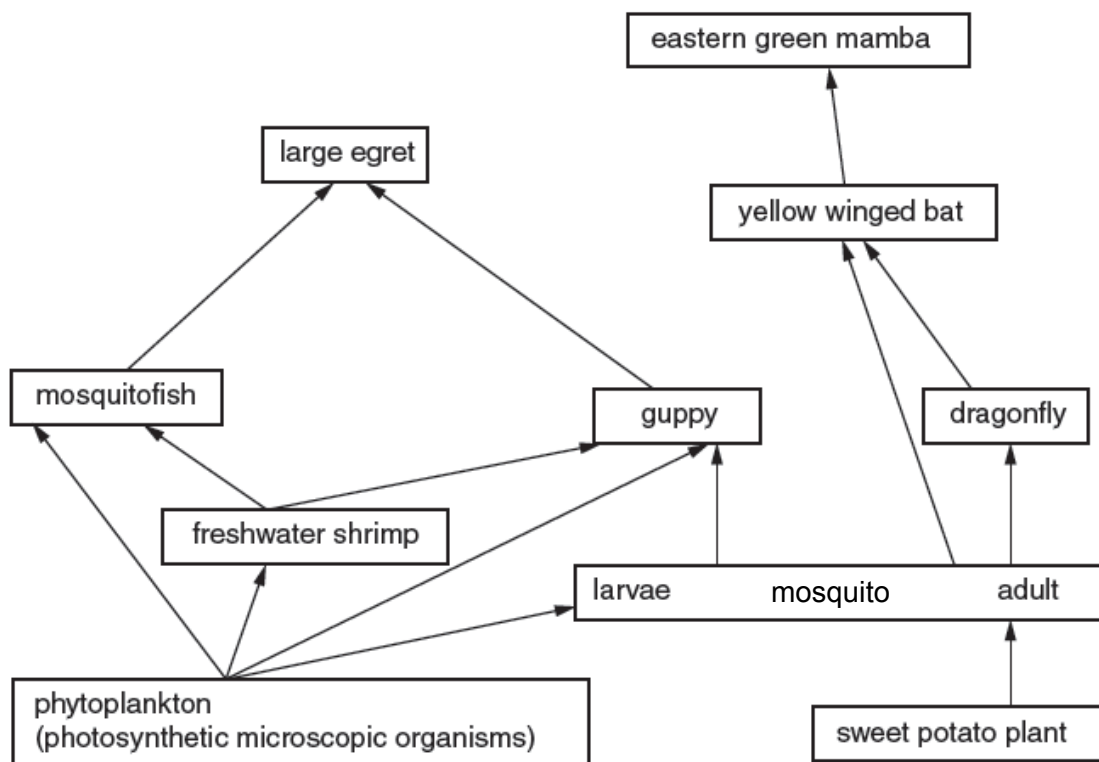


Fig. 3.9

- (ii) Explain which organism(s) found in this ecological niche will be greatly impacted by the release of *Wolbachia*-infected mosquitoes. [2]
1. Population of dragonflies decrease, as they only feeds on mosquitoes.
 2. Population of yellow-winged bats will also reduce as they feed on mosquitoes and dragonflies.
 3. Population of eastern green mamba will reduce as they feed on yellow-winged bats.

4. Population of sweet potato plant will increase as fewer mosquitoes will feed on them.

[Total: 28]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must set out in sections **(a)** and **(b)**, as indicated in the question.

EITHER

- 4 (a)** Describe the mechanisms that give rise to the vast diversity of antibodies prior to antigen stimulation and describe how the antibody is formed from mature mRNA. [14]

1. **Somatic recombination - maturing B cells differentiated from lymphoid stem cells.**
2. **Light chain and heavy chain in an antibody to be expressed,**
3. **Each light chain can only have one V and J segments,**
4. **Each heavy chain can only have one V, D, J segments.**
5. **The DNA segments recombined.**
6. **Different B cells have different genetic make-up for these loci.**
7. **Different combinations of V/D/J regions contribute to the antibody repertoire.**

8. **Junctional diversity - a variable number of nucleotides at the junction**
9. **Where the V and J segments are joined in light chains and**
10. **where the V, D and J segments are joined in heavy chains.**

11. **Combinatorial diversity - different heavy chains paired with different light chains to produce antibodies of different isotypes.**

12. **small ribosomal subunit, with tRNA binds to the 5' cap of the mRNA**

13. **The first aminoacyl-tRNA complex binds to the START codon (AUG) on the mRNA via complementary base pairing.**

14. **the large ribosomal subunit binds to the small ribosomal subunit to form the Translation Initiation Complex. The aminoacyl-tRNA complex is positioned at the P site of the ribosome.**

15. **The second aminoacyl-tRNA complex enters the A site and its anti-codon base pairs with the next codon of mRNA.**

16. **Peptidyl transferase catalyzes the formation of peptide bond between the first and second amino acids.**

17. **The ribosome moves along the mRNA to the next codon.**

18. **The first tRNA moves from the P site to the E site so that it can be released into the cytoplasm for reuse.**

19. **The second tRNA moves from the A site to the P site.**

20. **The third aminoacyl-tRNA complex enters the vacant A site**

21. and its anti-codon base pairs with the next codon of mRNA.
 22. The process of elongation is repeated until the ribosome reaches the STOP codon on the mRNA.
 23. Termination occurs when a STOP codon (UAA, UGA, UAG) of the mRNA is positioned at the A site.
 24. Release factor enters the ribosome and is positioned at the vacant A site and binds to the STOP codon.
 25. releases the polypeptide chain
 26. The polypeptide chain then fold into its secondary and tertiary structure.
 27. The polypeptide chain undergoes post-translational modification to add carbohydrates sidechains in GA.
- (b) Using a named disease, discuss how vaccination is an effective measure to control the disease. [11]
1. Smallpox
 2. Caused by *Variola minor* virus
 3. Infect Respiratory tract, Lymphatic system, Skin
 4. Prolonged direct face-to-face contact, Direct contact with infected bodily fluids (saliva droplets) or contaminated objects
 5. mass vaccination programme to achieve vaccination in
 6. 80% of the populations in each country,
 7. surveillance and prevention measures
 8. Infected individuals were easy to identify.
 9. Ring vaccination to reduced the chances of transmission and contained the disease.
 10. The live attenuated vaccine was used, elicits a strong immune response without the need of booster shots.
 11. Offers long-term or lifelong immunity/ protection
 12. Rapid and stronger immune response to combat infection in the future
 13. Activation of memory immune cells to produce antibodies to neutralize the pathogen and stop infection.
 14. Vaccine was cheap and easy to mass produce.
 15. Complete eradication through herd immunity and prevent epidemics and pandemics
 16. Herd immunity - it breaks the transmission cycle of the disease,
 17. Therefore those individuals who have NOT developed immunity are also protected from the disease.
 18. The virus did NOT infect animals, which made it easier to break the transmission cycle.

19. The virus did NOT mutate or change its surface antigens, therefore the same vaccine could be used anywhere in the world.
20. The vaccine was freeze-dried and thermostable. Thus, it has a longer shelf-life and could be kept at room temperature.

[Total: 25]

OR

- 5 (a) Describe the main stages of cell signaling and describe the roles of cAMP in eukaryotes and prokaryotes. [14]

1. Ligand has a specific shape that can bind to receptor
2. Induces a conformational change
3. activation of the receptor protein
4. initiates a signal transduction pathway/ cascade
5. protein phosphorylation
6. second messenger molecule
7. Changes in levels of enzyme activities in the cytoplasm
8. changes in the amount of specific proteins synthesized by the cell
9. Rearrangement of cytoskeleton
10. Signal is terminated by dissociation of ligand from receptor
11. Degradation/ Deactivation of ligand
12. (Any one example from GPCR / RTK)

13. cAMP is involved in the positive control of *lac* operon
14. cAMP binds to the allosteric site on CAP
15. binds to a CAP-binding site
16. bends DNA
17. easier for RNA polymerase to bind to promoter
18. *Lac* operon genes transcribed
19. β -galactosidase, *lac* permease and β -galactoside transacetylase

20. ligand binds to a GPCR
21. activated GPCR activates a G protein
22. activates the enzyme adenylyl cyclase
23. conversion of ATP to cAMP
24. cytosolic concentration of cAMP increase
25. cAMP acts as a second messenger
26. Activated protein kinase A initiates phosphorylation cascade,
27. eventually resulting in a cellular response
28. Signal is terminated when cAMP is degraded to AMP by phosphodiesterase

- (b) Compare the signaling pathways between G protein coupled receptor and receptor tyrosine kinase in relation to blood glucose regulation. [11]

Similarities

1. Ligands bind to the extracellular side
2. Termination occurs when the ligand is removed from the receptors

Differences

	Feature of comparison	GPCR	Receptor Tyrosine Kinase

3.	Ligands/ Signal molecules	Glucagon	Insulin
4.	Change to receptor upon ligand binding	Conformational change of the 7-helix transmembrane protein.	Dimerization of RTK.
5.	Chemical modifications of receptors	Absent.	Phosphorylation of tyrosine residues on the RKT subunit.
6.	Proteins associated with receptors	Causes a GTP molecule to displace the GDP molecule and activates the G protein.	Insulin response substrate (IRS) proteins binds to phosphorylated tyrosine residues on the receptor.
7.	Signal transduction	Activated adenylyl cyclase catalyses the conversion of ATP to cAMP.	Phosphorylated IRS proteins phosphorylate other relay proteins.
8.	Effect of second messengers / activated proteins	cAMP acts as a second messenger and activates intracellular proteins such as protein kinase A (PKA), which leads to a phosphorylation cascade.	Phosphorylated IRS proteins activate more than one signalling pathway.
9.	Types of cellular response	↓ glycogenesis	↑ glycogenesis ↑ protein synthesis ↑ lipogenesis ↓ gluconeogenesis
10.	Effect on blood glucose levels	Increased blood glucose levels	Decreased blood glucose levels
11.	Termination of signals	Protein phosphatases remove phosphate groups from the effector proteins	Endocytosis of the insulin- receptor complex.

[Total: 25]